Name
Date
Course

More Identities

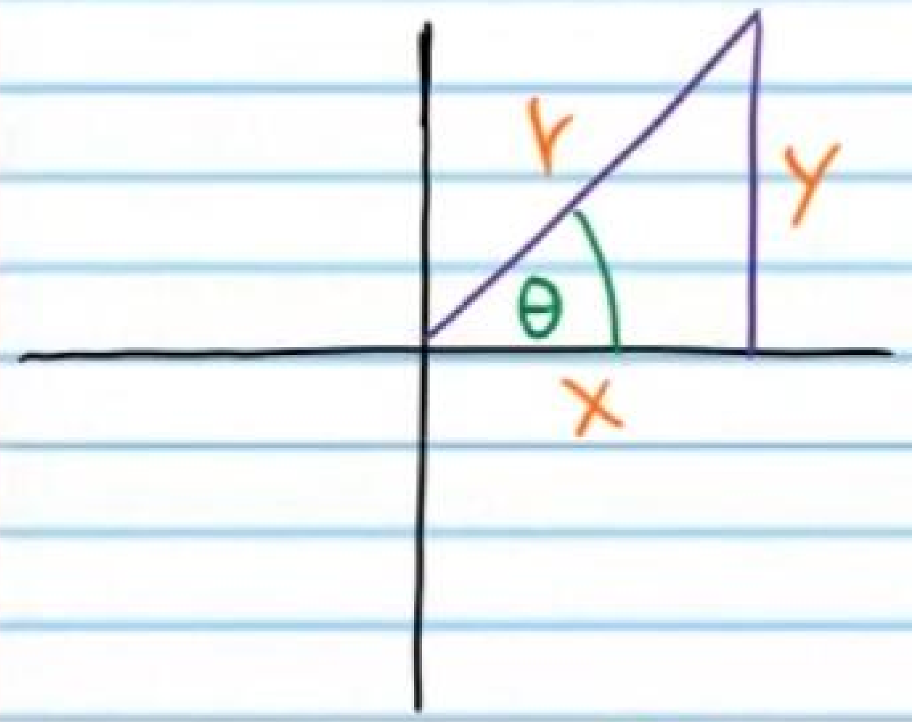
Opposite Angle Identities

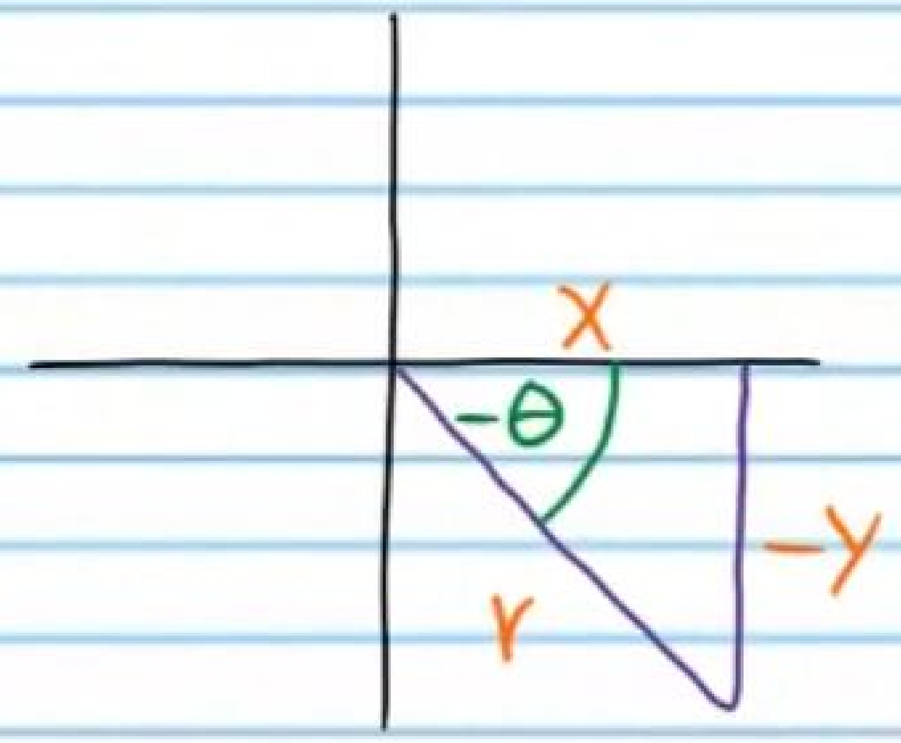
$\sin(-\theta) = -\sin\theta$ $\quad \cos(-\theta) = \cos\theta$ $\quad \tan(-\theta) = -\tan\theta$

$\csc(-\theta) = -\csc\theta$ $\quad \sec(-\theta) = \sec\theta$ $\quad \cot(-\theta) = -\cot\theta$

Short Explanations

Example I





$\sin\theta = \frac{y}{r}$ $\qquad \sin(-\theta) = \frac{-y}{r}$

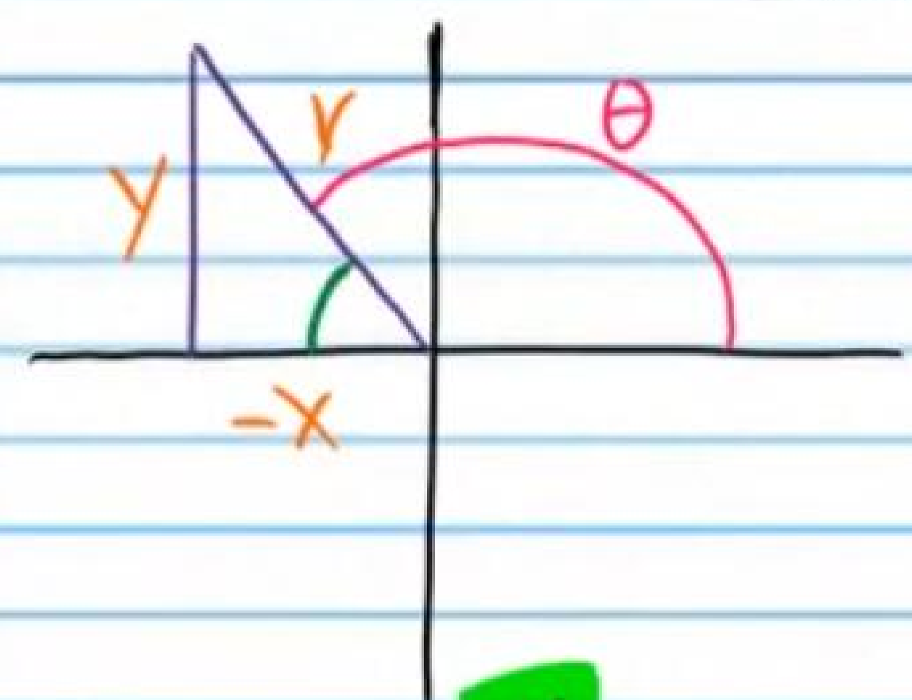
$\sin(-\theta) \neq \sin\theta$

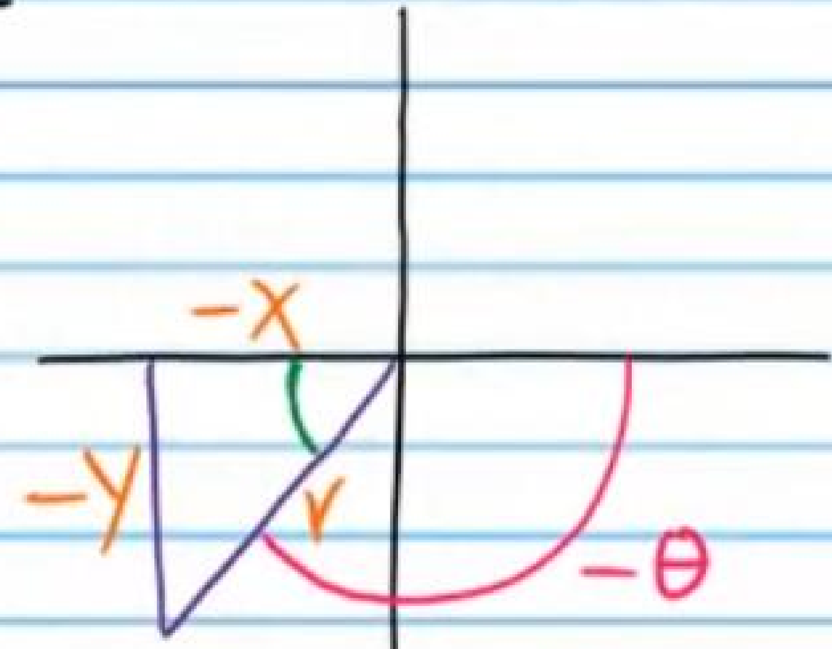
But

$\sin(-\theta) = -\sin\theta$

$-\frac{y}{r} = -\frac{y}{r}$

Example II



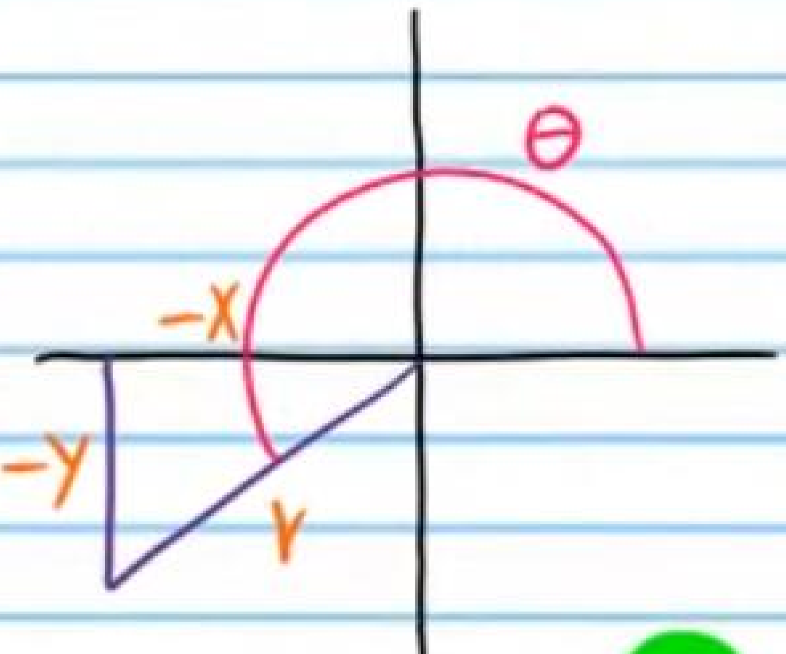


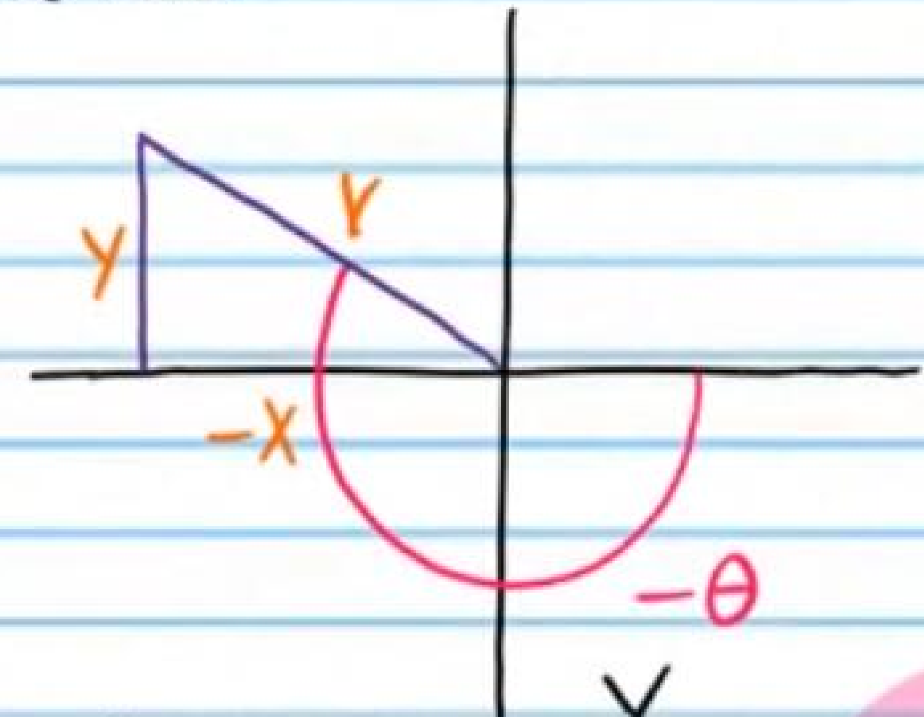
$\cos\theta = \frac{-x}{r}$ $\qquad \cos(-\theta) = \frac{-x}{r}$

$\cos(-\theta) = \cos\theta$

$-\frac{x}{r} = -\frac{x}{r}$

Example III





$\tan\theta = \frac{-y}{-x} = \frac{y}{x}$ $\qquad \tan(-\theta) = \frac{y}{-x} = -\frac{y}{x}$

$\tan(-\theta) \neq \tan\theta$

But

$\tan(-\theta) = -\tan\theta$

$-\frac{y}{x} = -\frac{y}{x}$

Practice

Prove that the following equations are true by converting the left side to the right.

① $\sin^2(-\theta) + \cos^2\theta = 1$

$(\sin(-\theta))^2 + \cos^2\theta =$

$(-\sin\theta)^2 + \cos^2\theta =$

$(-\sin\theta)(-\sin\theta) + \cos^2\theta =$

$\sin^2\theta + \cos^2\theta =$

$1 =$

② $\frac{\cos(-A)}{\sec(-A)} = \cos^2 A$

$\frac{\cos A}{\sec A} =$

$\cos A \div \sec A =$

$\cos A \div \frac{1}{\cos A} =$

$\cos A \cdot \frac{\cos A}{1} =$

$\cos^2 A =$

③ $-\cot\theta - \cot\theta = 2\cot(-\theta)$

$\cot(-\theta) + \cot(-\theta) =$

$2\cot(-\theta) =$

④ $\sec\alpha - \sec(-\alpha) = 0$

$\sec\alpha - \sec\alpha =$

$0 =$

⑤ $\sin(-x) + \sin x = 0$

$-\sin x + \sin x =$

$0 =$

⑥ $\frac{\tan(-\beta)}{\cot\beta} = -\tan^2\beta$

$\frac{-\tan\beta}{\cot\beta} =$

$-\tan\beta \div \cot\beta =$

$-\tan\beta \div \frac{1}{\tan\beta} =$

$-\tan\beta \cdot \tan\beta =$

$-\tan^2\beta =$

⑦ $2\cos\theta + \cos(-\theta) = 3\cos\theta$

$2\cos\theta + \cos\theta =$

$3\cos\theta =$

⑧ $\csc(-A)\csc(-A) = 1 + \cot^2 A$

$(-\csc A)(-\csc A) =$

$\csc^2 A =$

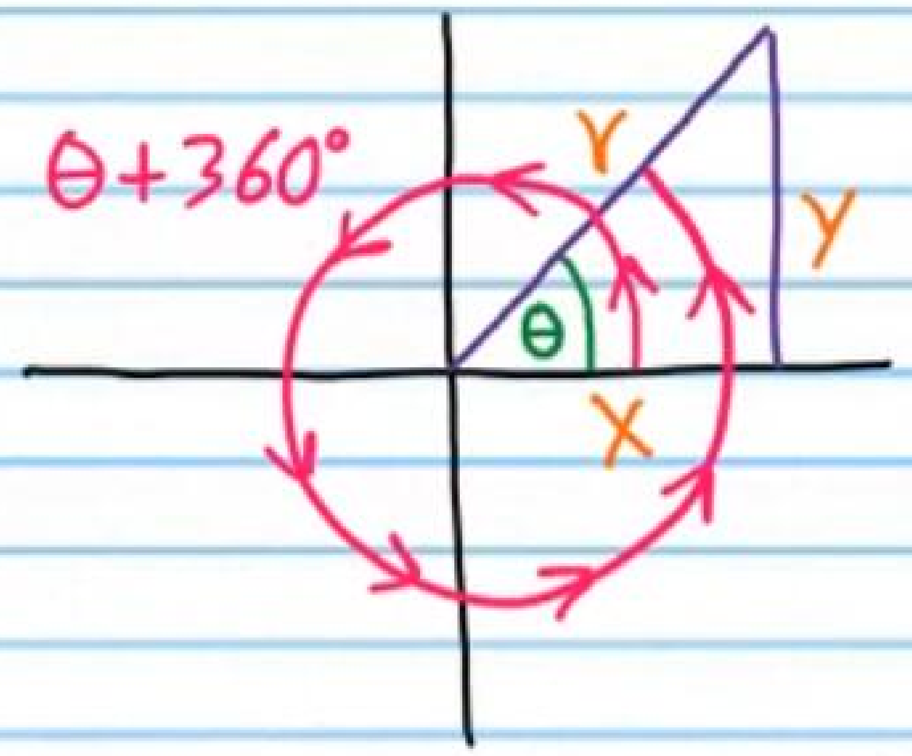
$1 + \cot^2 A =$

Complete Rotation and Half Rotation Identities

$\sin(\theta \pm 360^\circ) = \sin\theta$	$\sin(\theta \pm 180^\circ) = -\sin\theta$
$\cos(\theta \pm 360^\circ) = \cos\theta$	$\cos(\theta \pm 180^\circ) = -\cos\theta$
$\tan(\theta \pm 360^\circ) = \tan\theta$	$\csc(\theta \pm 180^\circ) = -\csc\theta$
$\csc(\theta \pm 360^\circ) = \csc\theta$	$\sec(\theta \pm 180^\circ) = -\sec\theta$
$\sec(\theta \pm 360^\circ) = \sec\theta$	$\tan(\theta \pm 180^\circ) = \tan\theta$
$\cot(\theta \pm 360^\circ) = \cot\theta$	$\cot(\theta \pm 180^\circ) = \cot\theta$

Short Explanations

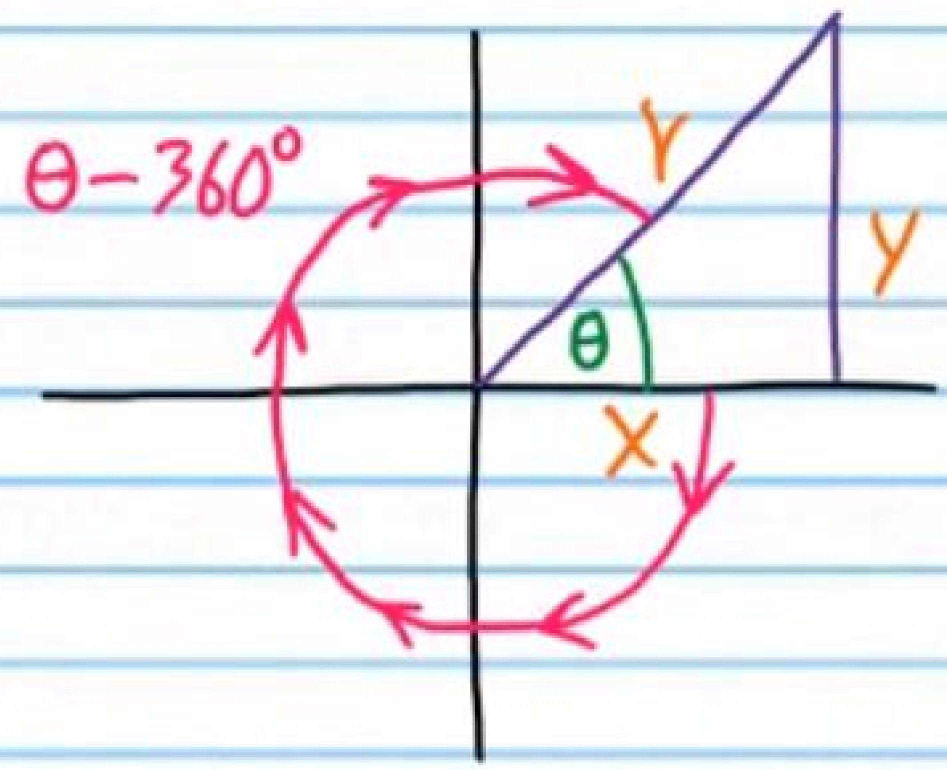
Example I



$\sin(\theta+360^\circ) = \sin\theta$

$\frac{y}{r} = \frac{y}{r}$

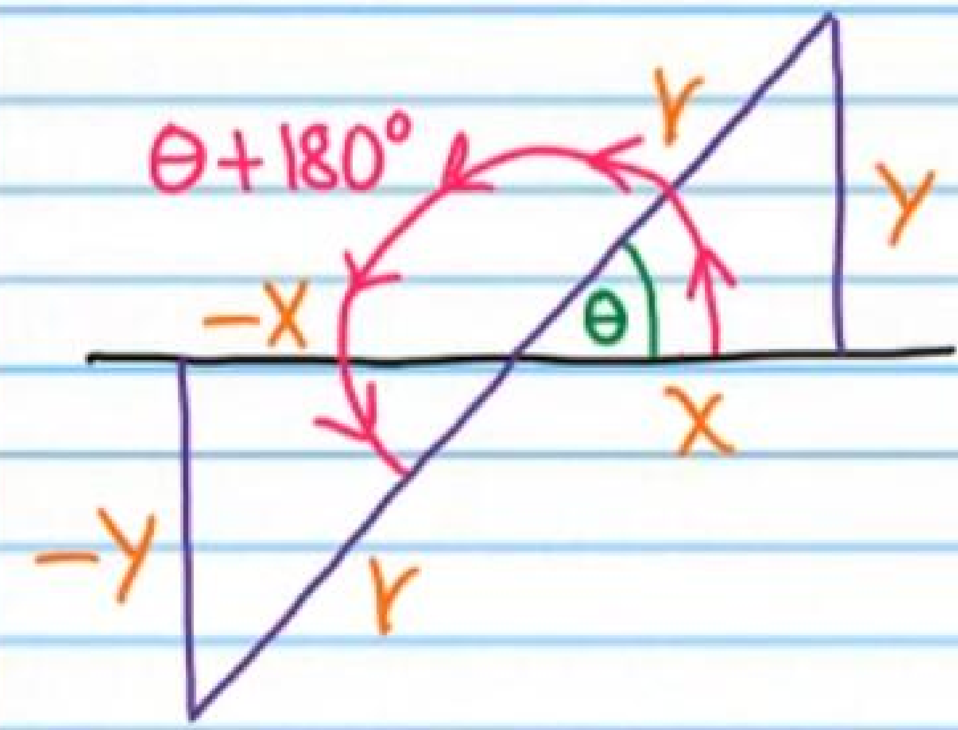
Example II



$\sec(\theta-360^\circ) = \sec\theta$

$\frac{r}{x} = \frac{r}{x}$

Example III



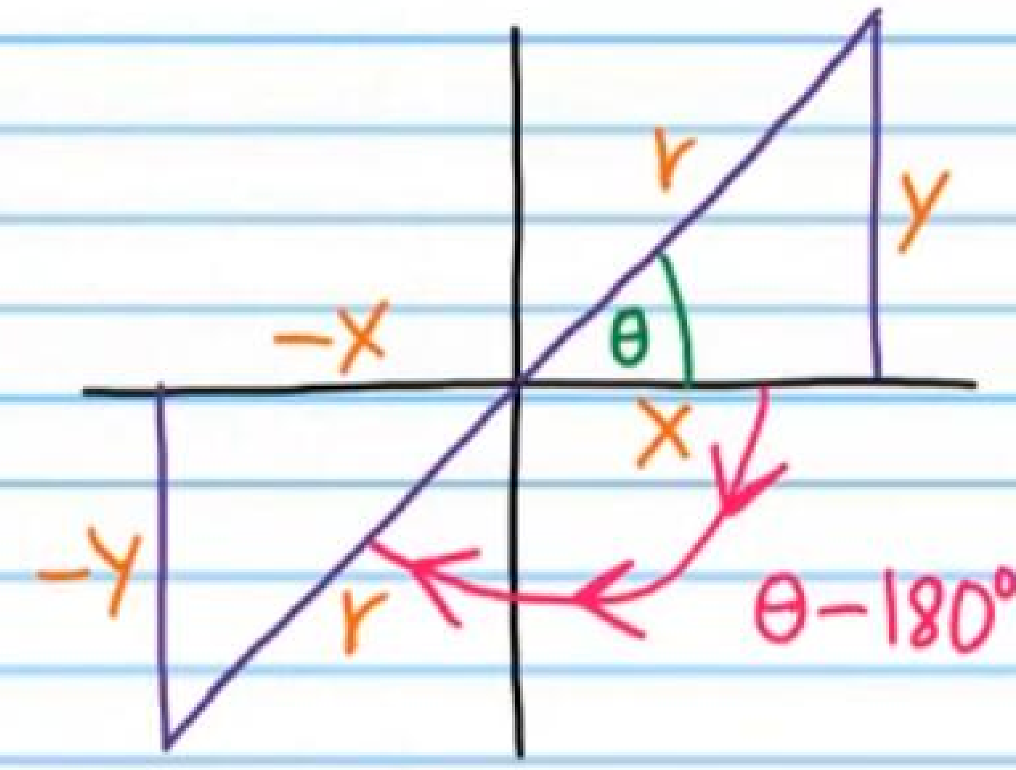
$\cos(\theta+180^\circ) \neq \cos\theta$

But

$\cos(\theta+180^\circ) = -\cos\theta$

$\frac{-x}{r} = -\frac{x}{r}$

Example IV



$\tan(\theta-180^\circ) = \tan\theta$

$\frac{-y}{-x} = \frac{y}{x}$

$\frac{y}{x} = \frac{y}{x}$

Practice

Prove that the following equations are true by converting the left side to the right.

⑨ $\frac{\csc(\theta-180^\circ)}{\csc\theta} = -1$

$\frac{-\csc\theta}{\csc\theta} =$

$-1 =$ ✓

⑩ $\tan A + 3\tan A = 4\tan(A+360^\circ)$

$4\tan A =$

$4\tan(A+360^\circ) =$ ✓

(11) $\dfrac{\cot(w+180)+\cot(w-360^\circ)}{\cot^2 w}=2\tan w$

$\dfrac{\cot w+\cot w}{\cot^2 w}=$

$\dfrac{2\cot w}{\cot^2 w}=$

$\dfrac{2\cot w}{\cot w\cot w}=$

$\dfrac{2}{\cot w}=$

$2\tan w=$

(12) $\sin(B-180^\circ)+\sin B=0$

$-\sin B+\sin B=$

$0\ =$

(13) $4\cos(\alpha+360^\circ)+2\cos(\alpha-360^\circ)=6\cos\alpha$

$4\cos\alpha+2\cos\alpha=$

$6\cos\alpha=$

(14) $\dfrac{\sec(\beta-180^\circ)}{\sec(\beta+180^\circ)}=1$

$\dfrac{-\sec\beta}{-\sec\beta}=$

$1\ =$

(15) $\dfrac{\tan(x+180^\circ)}{\tan^2(x-360^\circ)}=\cot x$

$\dfrac{\tan x}{\tan^2 x}=$

$\dfrac{\tan x}{\tan x\tan x}=$

$\dfrac{1}{\tan x}=$

$\cot x=$

(16) $\csc A-\csc(A-180^\circ)=2\csc(A-360^\circ)$

$\csc A-(-\csc A)=$

$\csc A+\csc A=$

$2\csc A=$

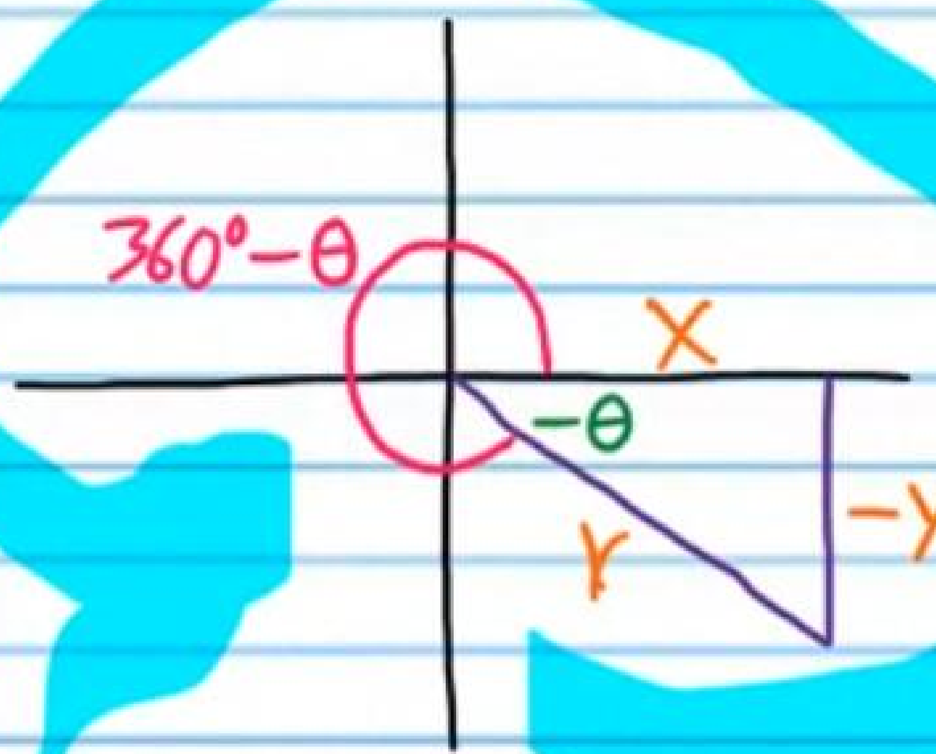
$2\csc(A-360^\circ)=$

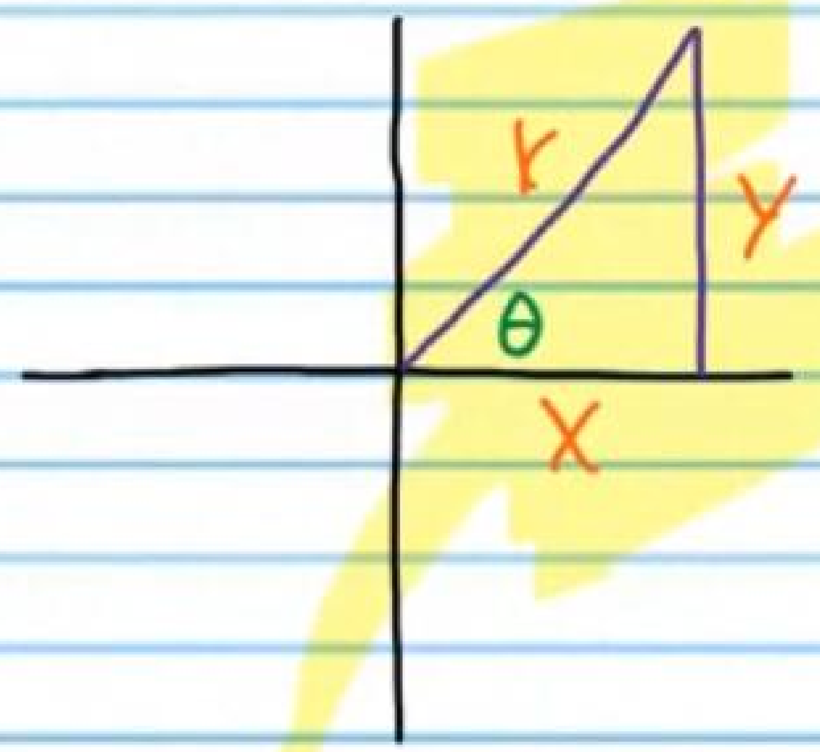
$360°-\theta$ Identities and $180°-\theta$ Identities

$\sin(360°-\theta)=-\sin\theta$	$\sin(180°-\theta)=\sin\theta$
$\cos(360°-\theta)=\cos\theta$	$\csc(180°-\theta)=\csc\theta$
$\tan(360°-\theta)=-\tan\theta$	$\cos(180°-\theta)=-\cos\theta$
$\csc(360°-\theta)=-\csc\theta$	$\sec(180°-\theta)=-\sec\theta$
$\sec(360°-\theta)=\sec\theta$	$\tan(180°-\theta)=-\tan\theta$
$\cot(360°-\theta)=-\cot\theta$	$\cot(180°-\theta)=-\cot\theta$

Short Explanations

Example I





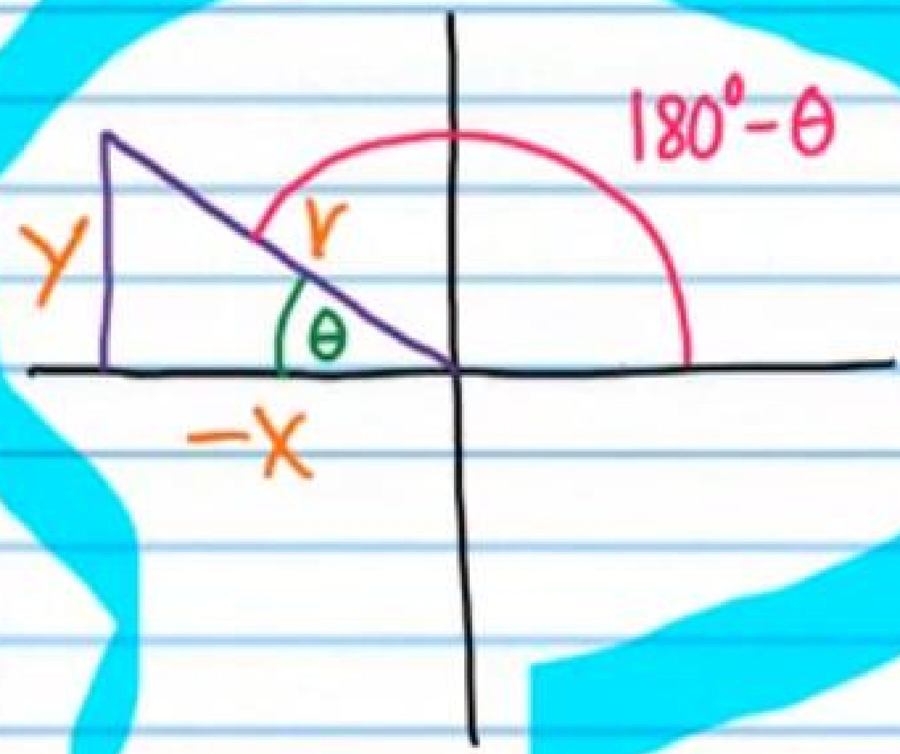
$\cot(360°-\theta)\neq\cot\theta$

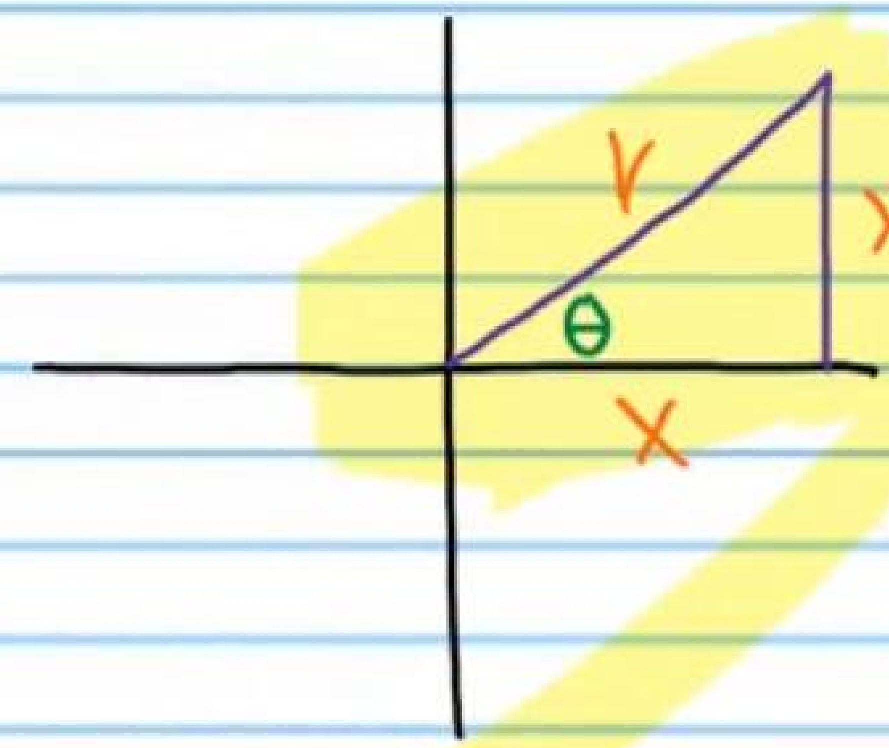
But

$\cot(360°-\theta)=-\cot\theta$

$$\frac{x}{-y}=-\frac{x}{y}$$

Example II





$\csc(180-\theta)=\csc\theta$

$$\frac{r}{y}=\frac{r}{y}$$

Practice

- (17) $\sin(360°-\theta)+2\sin\theta=\sin\theta$

  $-\sin\theta+2\sin\theta=$

  $\sin\theta=$

- (18) $\dfrac{\cos(360°-\beta)}{\cos(180°-\beta)}=-1$

  $\dfrac{\cos\beta}{-\cos\beta}=$

  $-1=$

(19) $4\sec A + 3\sec(180^\circ - A) = \sec(360^\circ - A)$

$4\sec A + 3(-\sec A) =$

$4\sec A - 3\sec A =$

$\sec A =$

$\sec(360^\circ - A) =$

(20) $\dfrac{\cos(360^\circ - \alpha)}{\sin(180^\circ - \alpha)} = \cot\alpha$

$\dfrac{\cos\alpha}{\sin\alpha} =$

$\cot\alpha =$

(21) $\dfrac{\sin(360^\circ - \beta)}{\cos(180^\circ - \beta)} = \tan\beta$

$\dfrac{-\sin\beta}{-\cos\beta} =$

$\tan\beta =$

(22) $\csc(360^\circ - w) + \csc w = 0$

$-\csc w + \csc w =$

$0 =$

(23) $\cot(180^\circ - x) - \cot(360^\circ - x) = 0$

$-\cot x - (-\cot x) =$

$-\cot x + \cot x =$

$0 =$

(24) $\sin(360^\circ - y) - \sin y = -2\sin(180^\circ - y)$

$-\sin y - \sin y =$

$-2\sin y =$

$-2\sin(180^\circ - y) =$

Quarter Rotation Identities

$\sin(\theta \pm 90^\circ) = \pm\cos\theta$ $\quad$ $\cos(\theta \pm 90^\circ) = \mp\sin\theta$

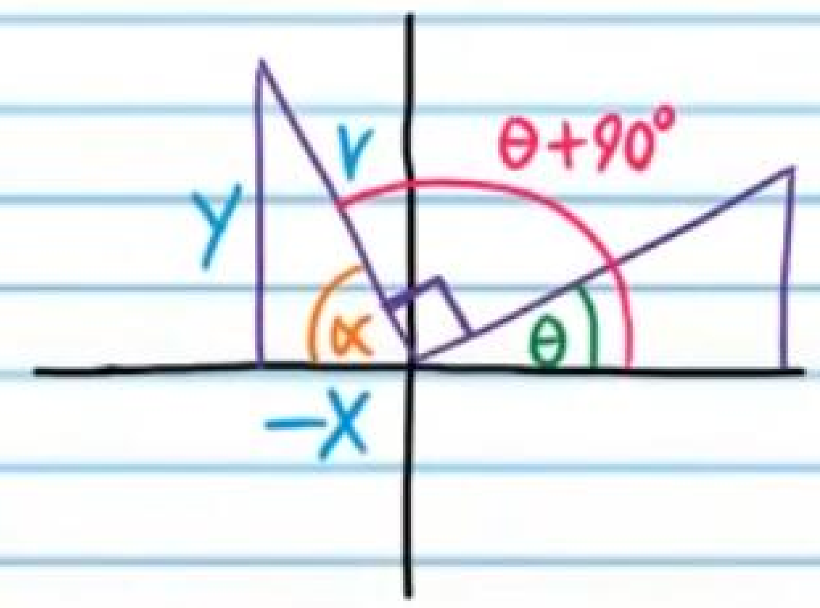
$\csc(\theta \pm 90^\circ) = \pm\sec\theta$ $\quad$ $\sec(\theta \pm 90^\circ) = \mp\csc\theta$

$\tan(\theta \pm 90^\circ) = -\cot\theta$

$\cot(\theta \pm 90^\circ) = -\tan\theta$

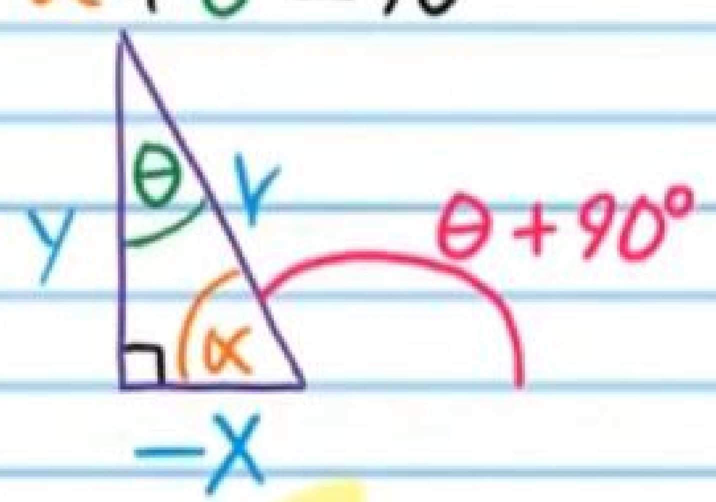
Short Explanations

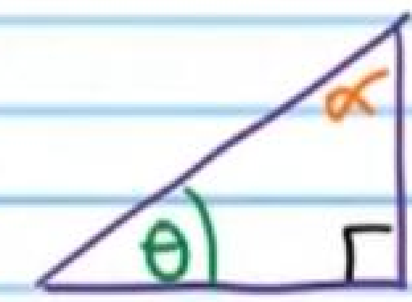
Example I



$\alpha + \theta + 90^\circ = 180^\circ$

$\alpha + \theta = 90^\circ$

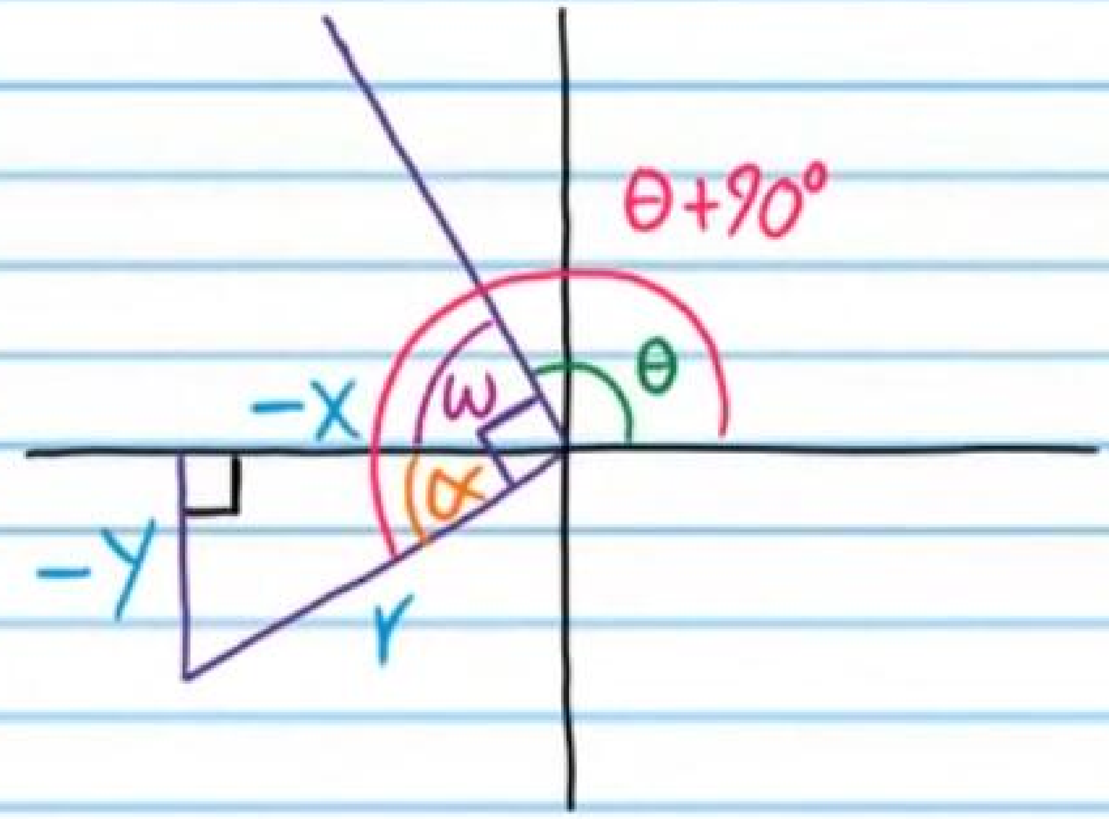




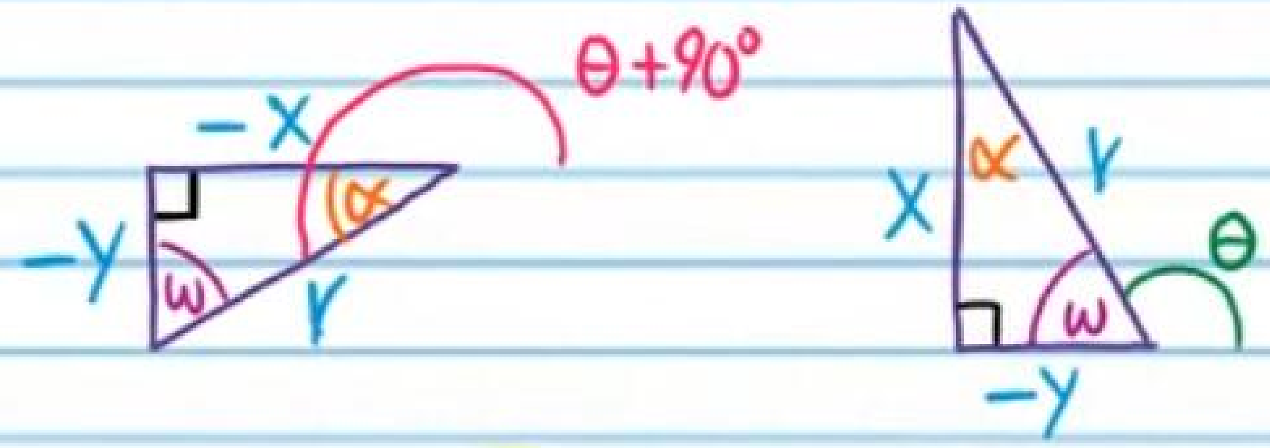
$\sin(\theta + 90^\circ) = \dfrac{y}{r}$, $\cos\theta = \dfrac{y}{r}$

$\sin(\theta + 90^\circ) = \cos\theta$

Example II



$\alpha + \omega = 90°$



$\sin(\theta+90°) = -\frac{y}{r}$, $\cos\theta = \cos\omega = -\frac{y}{r}$

$\sin(\theta+90°) = \cos\theta$

$90° - \theta$ Identities

$\sin(90°-\theta) = \cos\theta$ $\qquad$ $\csc(90°-\theta) = \sec\theta$

$\cos(90°-\theta) = \sin\theta$ $\qquad$ $\sec(90°-\theta) = \csc\theta$

$\tan(90°-\theta) = \cot\theta$

$\cot(90°-\theta) = \tan\theta$

Practice

(25) $\sin(\theta+90°) - \sin(\theta-90°) = 2\cos\theta$

$\cos\theta - (-\cos\theta) =$

$\cos\theta + \cos\theta =$

$2\cos\theta =$

(26) $\cot(A-90°)\tan A = -\tan^2 A$

$-\tan A \tan A =$

$-\tan^2 A =$

(27) $\cos^2(90°-B)\sec(90°-B) = \sin B$

$\sin^2 B \csc B =$

$\sin^2 B \frac{1}{\sin B} =$

$\frac{\sin^2 B}{\sin B} =$

$\frac{\sin B \sin B}{\sin B} =$

$\sin B =$

(28) $\csc^2(90°-W) = \tan^2 W + 1$

$\sec^2 W =$

$\tan^2 W + 1 =$

(29) $\cos(90° - \alpha) + \cos(\alpha + 90°) = 0$

$\sin\alpha + (-\sin\alpha) =$

$0 =$

(30) $\frac{\sin(x+90°)}{\cos(x-90°)} = \cot x$

$\frac{\cos x}{\sin x} =$

$\cot x =$

(31) $\cot^2(A - 90°) + 1 = \sec^2 A$

$(\cot(A-90°))^2 + 1 =$

$(-\tan A)^2 + 1 =$

$(-\tan A)(-\tan A) + 1 =$

$\tan^2 A + 1 =$

$\sec^2 A =$

(32) $\tan^3(90° - B)\cot B = \cot^4 B$

$\cot^3 B \cot B =$

$\cot^4 B =$

Using the Sum and Difference Identities to Derive the Previous Formulas

Use the sum and difference formulas to derive the following identities.

(33) $\sin(\theta + 90°) = \cos\theta$

$\sin(A+B) = \sin A \cos B + \sin B \cos A$

$\sin(\theta + 90°) = \sin\theta\cos 90° + \sin 90° \cos\theta$

$= (\sin\theta)(0) + (1)\cos\theta$

$= 0 + \cos\theta$

$= \cos\theta$

(34) $\sec(180° - x) = -\sec x$

$\sec(A - B) = \frac{1}{\cos(A-B)} = \frac{1}{\cos A \cos B + \sin A \sin B}$

$\sec(180° - x) = \frac{1}{\cos(180° - x)} = \frac{1}{\cos 180° \cos x + \sin 180° \sin x}$

$= \frac{1}{(-1)\cos x + (0)\sin x}$

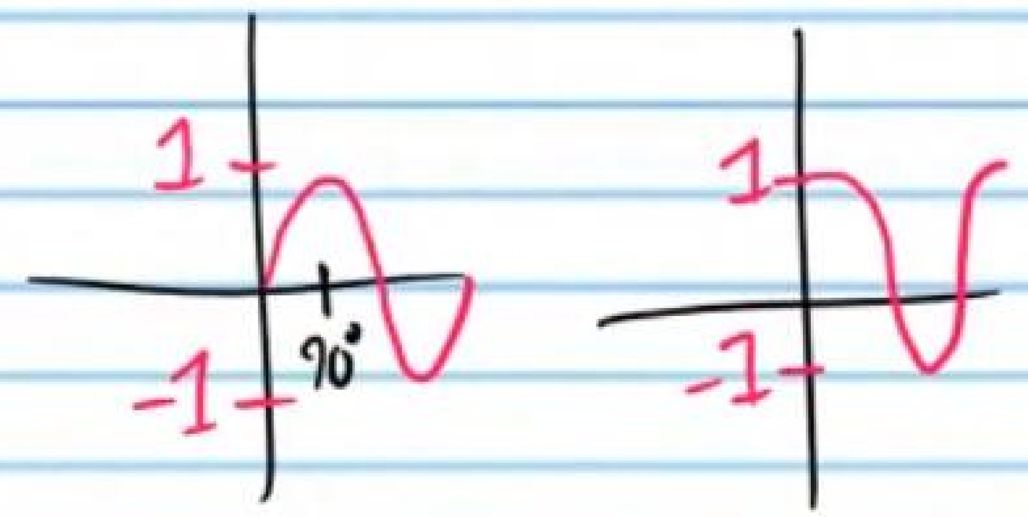
$= \frac{1}{-\cos x}$

$= -\sec x$

(35) $\tan(90° - W) = \cot W$

$$\tan(A-B) = \frac{\sin(A-B)}{\cos(A-B)} = \frac{\sin A\cos B - \sin B\cos A}{\cos A\cos B + \sin A\sin B}$$

$$\tan(90° - W) = \frac{\sin(90° - W)}{\cos(90° - W)} = \frac{\sin 90°\cos W - \sin W\cos 90°}{\cos 90°\cos W + \sin 90°\sin W}$$



$$= \frac{(1)\cos W - (\sin W)(0)}{(0)\cos W + (1)\sin W}$$

$$= \frac{\cos W - 0}{0 + \sin W}$$

$$= \frac{\cos W}{\sin W}$$

$$= \cot W$$

(36) $\cos(90° - \alpha) = \sin\alpha$

$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$

$$\cos(90° - \alpha) = \cos 90°\cos\alpha + \sin 90°\sin\alpha$$

$$= (0)(\cos\alpha) + (1)\sin\alpha$$

$$= 0 + \sin\alpha$$

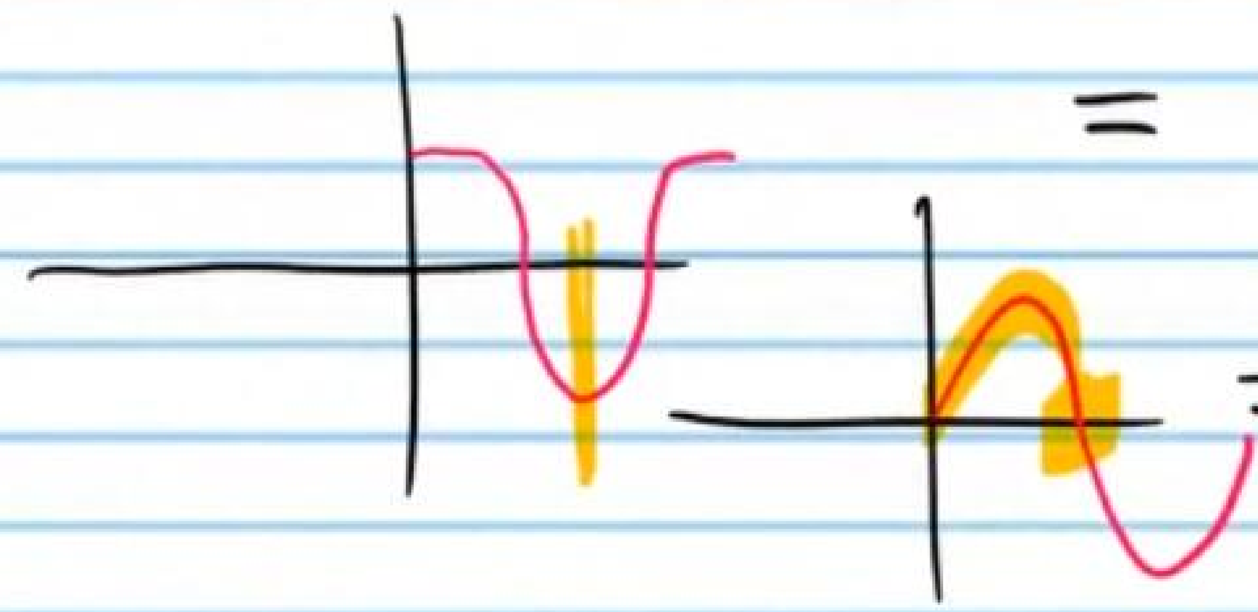
$$= \sin\alpha$$

(37) $\csc(\beta + 180°) = -\csc\beta$

$$\csc(A+B) = \frac{1}{\sin(A+B)} = \frac{1}{\sin A\cos B + \sin B\cos A}$$

$$\csc(\beta + 180°) = \frac{1}{\sin(\beta + 180°)} = \frac{1}{\sin\beta\cos 180° + \sin 180°\cos\beta}$$

$$= \frac{1}{(\sin\beta)(-1) + (0)\cos\beta}$$



$$= \frac{1}{-\sin\beta + 0}$$

$$= -\frac{1}{\sin\beta}$$

$$= -\csc\beta$$

(38) $\cot(\theta + 90°) = -\tan\theta$

$$\cot(A+B) = \frac{\cos(A+B)}{\sin(A+B)} = \frac{\cos A\cos B - \sin A\sin B}{\sin A\cos B + \sin B\cos A}$$

$$\cot(\theta + 90°) = \frac{\cos(\theta + 90°)}{\sin(\theta + 90°)} = \frac{\cos\theta\cos 90° - \sin\theta\sin 90°}{\sin\theta\cos 90° + \sin 90°\cos\theta}$$

$$= \frac{(\cos\theta)(0) - (\sin\theta)(1)}{(\sin\theta)(0) + (1)(\cos\theta)}$$

$$= \frac{-\sin\theta}{\cos\theta}$$

$$= -\tan\theta$$

Product to Sum Formulas

$$\sin A\cos B = \frac{1}{2}\left[\sin(A+B)+\sin(A-B)\right]$$

$$\cos A\sin B = \frac{1}{2}\left[\sin(A+B)-\sin(A-B)\right]$$

$$\cos A\cos B = \frac{1}{2}\left[\cos(A+B)+\cos(A-B)\right]$$

$$\sin A\sin B = \frac{1}{2}\left[\cos(A-B)-\cos(A+B)\right]$$

A Proof of the First Equation

$$\sin(A+B) = \sin A\cos B + \sin B\cos A$$

$$\sin(A-B) = \sin A\cos B - \sin B\cos A$$

$$\sin(A+B) + \sin(A-B) = \sin A\cos B + \sin B\cos A + \sin A\cos B - \sin B\cos A$$

$$\sin(A+B)+\sin(A-B) = 2\sin A\cos B$$

$$\frac{1}{2}\left[\sin(A+B)+\sin(A-B)\right] = \sin A\cos B$$

Sum to Product Formulas

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin A - \sin B = 2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$$

$$\cos A + \cos B = 2\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$$

$$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

A Proof of the First Equation

$$\sin(A+B)+\sin(A-B) = 2\sin A\cos B$$

Replace A with $\frac{A+B}{2}$ and B with $\frac{A-B}{2}$.

$$\sin\left(\frac{A+B}{2}+\frac{A-B}{2}\right)+\sin\left(\frac{A+B}{2}-\frac{A-B}{2}\right) = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin\left(\frac{A+A+B-B}{2}\right)+\sin\left(\frac{A+B-A+B}{2}\right) = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin\left(\frac{2A}{2}\right)+\sin\left(\frac{2B}{2}\right) = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$